

(Q)  $Y = (A \oplus B)(B \odot C) \cdot C$

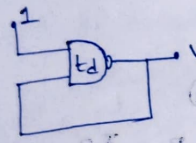
Which combination of (A, B, C) will make  $Y = 1$ ?

(a) 1, 0, 1 (b) 0, 0, 1 (c) 1, 1, 1 (d) 0, 1, 1

A.  $\oplus \rightarrow A \& B$  shouldn't be same

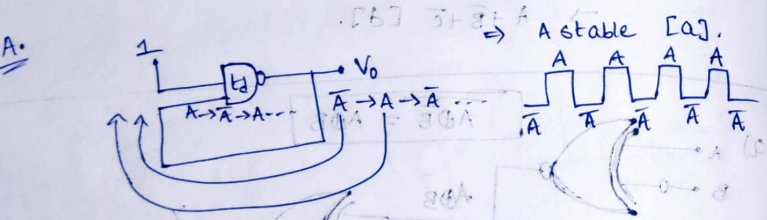
(d)  $(0 \oplus 1) \cdot (1 \odot 1) \cdot 1 = 1$

(Q) The shown NAND gate has a finite (Non-zero) delay ( $t_d$ ).



The circuit acts as

- (a) A stable Multivibrator
- (b) Bi stable
- (c) Mono stable
- (d) None



(Q)  $\overline{B}C + AB + A\overline{C}$  equals to

- (a)  $[A + \overline{B}][\overline{A} + C]$
- (b)  $[A + \overline{B}][A + C]$
- (c)  $A + B$
- (d) None

$Y = AB + AC + \overline{B}C$

$= \overline{B}C + A[B + C]$

$= \overline{B}C + A[\overline{B}C]$

$= X + \overline{X}Y = X + Y = \overline{B}C + A$

$= A + \overline{B}C$

$= [A + \overline{B}][A + C]$  [C].

(Q)  $A \oplus (\overline{A} + B)$  equals

- (a) 1 (b) 0 (c)  $\overline{A}\overline{B}$  (d) None

A.  $A \oplus (\overline{A} + B) = \overline{A} \cdot \overline{A} \cdot \overline{B} + A \cdot \overline{A} \cdot \overline{B} + A \cdot \overline{A} \cdot B + A \cdot A \cdot \overline{B} + A \cdot A \cdot B$

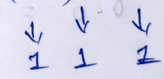
$\overline{A}B + A + \overline{A}B = \overline{A}B + A$

(Q) Assuming that the variables and their complements are available, then minimum number of 2-input NOR gate needed to implement 2 input XOR gate is 3?

A.  $Y = \overline{A}B + A\overline{B} = (\overline{A} + \overline{B}) \cdot (A + B)$

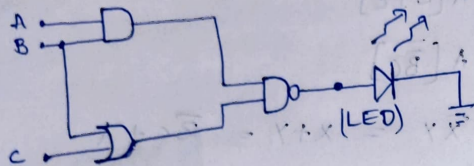
A, B,  $\overline{A}$ ,  $\overline{B}$   $\rightarrow$  Available

$(C + D) \cdot (A + B) \rightarrow 3$



MCQ

(Q)



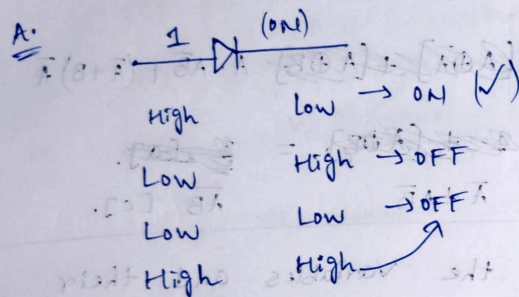
Possible Combination of A, B, C to turn on LED.

(a)  $A=0, B=0, C=1$

(b)  $A=1, B=0, C=0$

(c)  $A=1, B=1, C=0$

(d)  $A=1, B=1, C=1$



(a)  $A \cdot B = 0 \cdot 0 = 0$   
 $B + C = 0 + 1 = 1$   
 $0 \cdot 1 = 0$

(b)  $A \cdot B = 1 \cdot 1 = 1$   
 $B + C = 1 + 0 = 1$   
 $1 \cdot 1 = 1$

(c)  $A \cdot B = 1 \cdot 0 = 0$   
 $B + C = 0 + 0 = 0$   
 $0 \cdot 0 = 0$

(d)  $A \cdot B = 1 \cdot 1 = 1$   
 $B + C = 1 + 1 = 1$   
 $1 \cdot 1 = 1$

(Q) If  $A \oplus B = C$ ,  $\overline{B} \oplus C$  will be?

(a) A (b)  $\overline{A}$  (c) AB (d) 1

A:  $\overline{B} \oplus C = \overline{B \oplus C} = \overline{A}$  [b]

(Q)  $\overline{A \oplus B \oplus C}$  equals to

(a)  $A \oplus B \oplus C$  (b)  $A \oplus B \oplus C$  (c)  $A \oplus B \oplus C$  (d) None

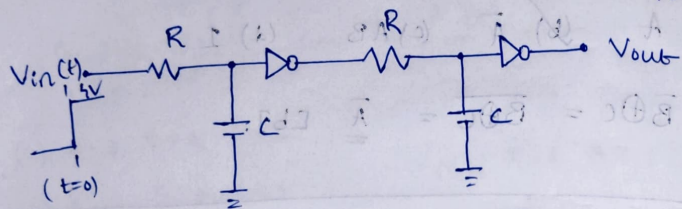
A:  $\overline{A \oplus B \oplus C} = \overline{A \oplus B \oplus C} = \overline{A \oplus B \oplus C}$

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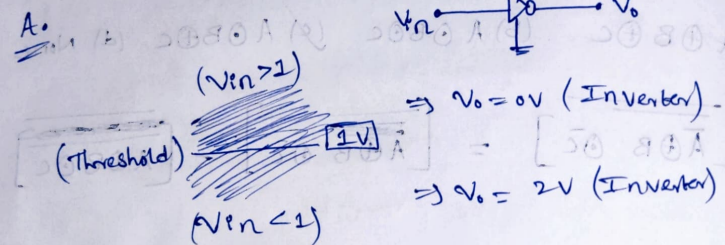


(Q) The inverters are driven off a supply of 2V and their threshold points are at mid supply. At what point of time  $V_o(t)$  goes to  $V_{DD}$ ?



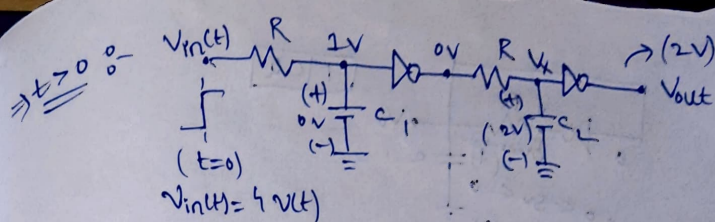
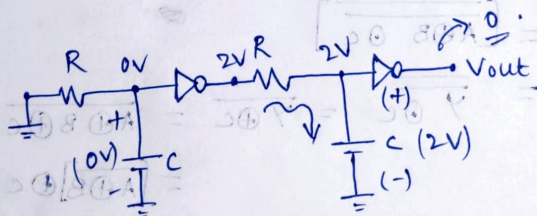
$$V_{in}(t) = 2V(t)$$

$$RC = 1 \text{ Sec.}$$



$$\Rightarrow V_T = 2/2 = 1V$$

$t < 0$ :



When  $V_{in}$  goes down to  $0.999999999V$

$V_{out}$  goes to  $2V$ .

$\Rightarrow$  To discharge  $C_2$ , you need 1st inverter OP as 0. To get 0, its input should be  $1V$ , i.e.,  $C_1$  should be charged to  $1V$ .

$$\Rightarrow 1 [1 - e^{-t/1}] = 1 \Rightarrow t_1 = \ln(2/1) \text{ Sec (charging } C_1)$$

$$\Rightarrow 2 e^{-t_2/1} = 1 \Rightarrow t_2 = \ln(2/1) + \ln(2/1) \text{ Sec}$$

$$\Rightarrow t_2 = \ln(8/3) \text{ Sec.}$$

Which of the statement is true?

- (a) The CKT works as a Astable Multivibrator.  
 (b) " " " Mono Stable " "  
 (c) " " " Bi Stable " "  
 (d) The CKT produces a Periodic sine wave.  
 (e) if RC is low.

A. It is clearly having No stable state.  $[A, D]:$

Boolean Fn.

## Basics of Number System :

### ① Decimal Number System :-

[Base 10] = 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

Eg:-  $729 = 7 \times 10^2 + 2 \times 10^1 + 9 \times 10^0$   
 MSB (Most Significant Digit)      LSB (Least Significant Digit)

## ② Binary Numbers System :-

Base 2 [Base 2] = 0, 1  $\rightarrow$  (bits)

Eg:-  $110 = 0 \times 2^0 + 1 \times 2^1 + 1 \times 2^2 \rightarrow (\text{LSB})$

$$2 + 4 = 6$$

MSB - Most Significant bit

LSB - Least Significant Bit

$$\text{Eq: } 1 \rightarrow 401 = 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 1 + 0 + 1 = 2.$$

$$\rightarrow 111 = 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 1 + 2 + 1 = 7.$$

$$\rightarrow 1101 = 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$
$$= 1 + 4 + 8 = \underline{13}.$$



$$\rightarrow 111001 = 1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$= 1 + 8 + 16 + 32 = 57$$

$\Rightarrow$  So, the representation is

$\Rightarrow$  2-Variables :-

$$\rightarrow 00 = 0 \times 2^0 + 0 \times 2^1 = 0$$

$$\rightarrow 01 = 0 \times 2^0 + 1 \times 2^1 = 1$$

$$\rightarrow 10 = 1 \times 2^0 + 0 \times 2^1 = 2$$

$$\rightarrow 11 = 1 \times 2^0 + 1 \times 2^1 = 3$$

| (Decimal) | (MSB) | (LSB) |
|-----------|-------|-------|
| D         | $b_2$ | $b_1$ |
| 0         | 0     | 0     |
| 1         | 0     | 1     |
| 2         | 1     | 0     |
| 3         | 1     | 1     |

★ For  $n$ -Variables  $\Rightarrow 2^n - 1 = \text{Max. decimal}$   
 No. of Possibilities

$\Rightarrow$  Product term :- Product of Literals.

Eg:-  $AB, \bar{A}B, A\bar{B}, ABC, A\bar{B}C, ABCD$

$\Rightarrow$  Sum term :- Sum of Literals.

Eg:-  $A+B, \bar{A}+B, A+\bar{B}, \bar{A}+\bar{B}+\bar{C}, \bar{A}+B+\bar{C}, A+B+\bar{C}+D$

$\Rightarrow$  Minterm & Maxterm :-

• Minterm :- A logical AND operation including all Variables, where each Variable can be itself (or) its Complement.

$\rightarrow$  op of minterm Function is 1.

$\rightarrow$  Represented by "m".

$\rightarrow 1 \rightarrow \text{Normal form} ; 0 \rightarrow \text{Complemented Form}$

Eg:-

| D | A | B | Minterm                 | Expression |
|---|---|---|-------------------------|------------|
| 0 | 0 | 0 | $\bar{A} \cdot \bar{B}$ | $m_0$      |
| 1 | 0 | 1 | $\bar{A} \cdot B$       | $m_1$      |
| 2 | 1 | 0 | $A \cdot \bar{B}$       | $m_2$      |
| 3 | 1 | 1 | $A \cdot B$             | $m_3$      |

(2-Variable)

Eg:-  $f(W, X, Y, Z) = \bar{W} X Y \bar{Z}$  Expression for Minterm?

$$\rightarrow \bar{W} \cdot X \cdot Y \cdot \bar{Z} = 01110$$

$$= 0 \times 2^0 + 1 \times 2^1 + 1 \times 2^2 + 0 \times 2^3$$

$$= 2 + 4 = 6 \text{ (Minterm - } m_6 \text{)}$$

(★)  $n$ -Variable =  $2^n$  min terms